

AI-Powered Cognitive Screening & Dementia Risk Analysis Platform

Himesha Senaratne

Department of Software Engineering
Sri Lanka Institute of Information
Technology (SLIIT)
Malabe, Sri Lanka
IT22358998@my.sliit.lk

Piruthivi Kulasegaram

Department of Software Engineering
Sri Lanka Institute of Information
Technology (SLIIT)
Malabe, Sri Lanka
IT22358820@my.sliit.lk

Natheer Hashmath Al-hazan

Department of Software Engineering
Sri Lanka Institute of Information
Technology (SLIIT)
Malabe, Sri Lanka
IT22888952@my.sliit.lk

Malindi Dissanayake

Department of Software Engineering
Sri Lanka Institute of Information
Technology (SLIIT)
Malabe, Sri Lanka
IT22901644@my.sliit.lk

Ms. Manori Gamage

Department of Information Technology
Sri Lanka Institute of Information
Technology (SLIIT)
Malabe, Sri Lanka
manori.g@sliit.lk

Ms. Kaushalya Rajapakse

Department of Software Engineering
Sri Lanka Institute of Information
Technology (SLIIT)
Malabe, Sri Lanka
kaushalya.r@sliit.lk

Abstract— Early identification of dementia risk is essential for timely clinical intervention; however, traditional screening approaches often rely on manual assessments and limited behavioral indicators. This study presents CogniCare, a multimodal AI-assisted cognitive screening platform designed to support clinicians through integrated digital assessments. The system enables users to complete lifestyle profiling, cognitive and functional evaluations, handwriting analysis, and guided speech tasks through a unified platform. Collected health and behavioral data are analyzed using machine learning and standardized assessment methods to estimate dementia risk, while the results are integrated through rule-based analysis to generate an initial screening outcome. Speech and handwriting modules further analyze behavioral patterns associated with cognitive decline, and all outputs are consolidated into a comprehensive report available through a clinician dashboard. Guided by DSM-5 diagnostic criteria, physicians determine whether additional neurological evaluation is required, and MRI scans can be further analyzed using deep learning-based dementia stage classification to assist clinical decision-making. The proposed platform demonstrates the potential of integrating multimodal digital assessments with clinical workflows for scalable dementia risk screening while ensuring that final diagnostic decisions remain under medical supervision. Experimental evaluation demonstrates strong performance across system modules, where the wearable-based CatBoost model achieved AUC 0.989 and 94.9% accuracy, the speech hybrid model achieved F1-score 0.82, the handwriting model achieved F1-score 0.870, and the MRI ensemble model achieved 93.75% accuracy with Macro F1-score 0.9606.

Keywords—Dementia Screening, Multimodal Health Analysis, Cognitive Assessment, Machine Learning, Speech Analysis, Handwriting Analysis, MRI Classification, Clinical Decision Support.

I. INTRODUCTION

A. Background

Dementia is a progressive neurological disorder that affects cognitive functions such as memory, language, reasoning, and the ability to perform everyday activities. [1] It is one of the leading causes of disability and dependency among older adults worldwide. The number of individuals living with dementia is rapidly increasing due to ageing populations, placing significant pressure on healthcare systems.

Early detection of cognitive decline is critical because timely interventions can slow disease progression, improve patient management, and enhance quality of life. [2] Traditional dementia diagnosis often relies on clinical assessments, cognitive screening tests, and neuroimaging techniques conducted in specialized healthcare settings. [3] Common screening tools such as cognitive assessments and functional ability evaluations provide valuable insights into memory, language, and executive functioning.

However, these assessments are typically conducted periodically in clinical environments and require trained medical professionals. As a result, they may not capture subtle behavioral and physiological changes that occur in the early stages of cognitive decline. Recent advances in artificial intelligence, wearable health technologies, and digital behavioral analysis offer new opportunities for developing automated systems capable of supporting early dementia screening using multiple data sources.

Behavioral indicators such as speech patterns, handwriting dynamics, and daily activity levels [4], together with medical imaging data such as brain MRI scans, can provide important signals related to cognitive health [5] [6]. By integrating these heterogeneous data sources into a unified analytical framework, intelligent systems can assist clinicians in identifying early signs of cognitive impairment more effectively.

B. Problem Statement

Despite progress in dementia research, several limitations exist in current screening and monitoring approaches. Traditional cognitive assessments are often manual and require face-to-face clinical evaluations, making them time-consuming and difficult to scale for large populations. These tests are typically performed at specific time intervals and therefore do not support continuous monitoring of cognitive health.

Additionally, many existing screening methods rely on single data modalities. Some approaches focus exclusively on neuroimaging data, while others analyze cognitive test results or behavioral patterns independently. This fragmented approach limits the ability to capture the complex and multidimensional nature of cognitive decline.

Furthermore, lifestyle and physiological factors such as physical activity, sleep patterns, and mental health indicators are increasingly recognized as important contributors to dementia risk. However, many existing systems do not integrate wearable health data or behavioral indicators into the screening process. The lack of automated systems capable of integrating behavioral, physiological, and neuroimaging data limits the development of scalable and accessible dementia screening solutions.

C. Research Gap

Existing research on dementia detection often focuses on isolated diagnostic approaches. Some systems utilize machine learning models to analyze brain MRI scans for structural abnormalities associated with dementia. Other studies investigate behavioral indicators such as speech characteristics or handwriting patterns to identify cognitive impairment. Similarly, wearable health monitoring systems have been developed to track lifestyle factors that may influence neurological health.

However, these approaches are developed independently and rarely integrated into a comprehensive screening platform. The absence of a multimodal framework that combines behavioral signals, physiological data, and neuroimaging analysis limits the effectiveness of current digital dementia screening systems.

An integrated system capable of combining handwriting analysis, speech-based cognitive screening, wearable health monitoring, and MRI-based neuroimaging analysis could provide a more holistic understanding of cognitive health. Such a multimodal approach has the potential to improve screening accuracy, enable earlier detection of cognitive decline, and provide clinicians with more interpretable insights for decision-making.

D. Objective

The objective of this research is to design and develop CogniCare, an AI-powered multimodal cognitive screening and dementia risk analysis platform that integrates handwriting behavior analysis, wearable health and lifestyle monitoring, speech-based cognitive assessment, and MRI-based neuroimaging classification.

The proposed system collects and analyzes data from multiple sources, including digital handwriting inputs, speech recordings from guided tasks, wearable device data obtained through health monitoring APIs, and brain MRI images. Machine learning and deep learning techniques are applied to extract meaningful features from each modality and generate predictions related to dementia risk and disease stage.

By integrating these diverse data sources into a unified analytical framework, the platform aims to support early dementia detection, provide interpretable decision-support insights for clinicians, and enable scalable cognitive health monitoring in both clinical and home-based environments.

II. RELATED WORK

Digital cognitive screening tools improve the reliability of traditional paper-based assessments. Newman et al. [7] introduced ACE mobile, a tablet-based version of ACE-III that reduces scoring errors through automated scoring and standardized administration. This demonstrates the benefits of computerized cognitive assessments in improving diagnostic consistency.

A. Smart Health Monitoring & Dementia Screening Risk Analysis

Wearable devices [8] enable continuous monitoring of physiological signals, activity, and sleep behavior associated with dementia risk. Albers et al. [9] used wearable sensors and machine learning to detect behavioral symptoms of dementia, while Doherty et al. [10] found that walking speed, physical activity, and sleep patterns are significantly associated with dementia risk.

B. Handwriting and Drawing-Based Assessment

Handwriting and drawing tasks have been explored as digital biomarkers for cognitive decline. Araújo et al. [11] demonstrated that handwriting dynamics can reflect cognitive and motor impairments. Zhang et al. [12] and Drotár et al. [13] further showed that temporal and kinematic handwriting features can support machine learning-based detection of neurodegenerative diseases.

C. Dementia screening through guided speech-task analysis

Speech and language changes are recognized as early indicators of cognitive decline. De Paula et al. [14] applied automated speech analysis to identify linguistic patterns associated with Alzheimer's disease. Luz et al. [15] introduced the ADReSS dataset for dementia detection from speech recordings, while König et al. [16] confirmed strong correlations between automated speech features and clinician-rated impairments.

D. Dementia stage classification using MRIs

Structural MRI is widely used for automated dementia detection. Bäckström et al. [17] developed a large-scale deep learning classifier using multiple MRI datasets. Salvatore et al. [18] and Abdullah et al. [19] demonstrated that machine learning and deep learning models can effectively distinguish cognitively normal individuals from patients with mild cognitive impairment and Alzheimer's disease. Inspired by these studies, the proposed system applies an ensemble of CNN models for MRI-based dementia stage classification with explainable AI techniques.

III. METHODOLOGY

A. Smart Health Monitoring & Dementia Screening Risk Analysis

The Smart Health Monitoring module collects multimodal health, behavioral, and assessment data through the CogniCare application. Demographic and lifestyle information includes age, gender, education level, dominant hand, family history of dementia, body weight, body temperature, smoking status, alcohol consumption, diet type, chronic health conditions, diabetes status, sleep behavior, and PHQ-2 depression indicators. Physiological data [20] such as heart rate, step count, and physical activity are obtained through wearable device integration using Google Fit. Cognitive performance is evaluated using ACE-III [21] tasks, while functional ability is assessed using the Lawton Instrumental Activities of Daily Living (IADL) scale [22]. The collected data undergoes preprocessing, including data cleaning, removal of irrelevant features, handling missing values, and normalization, while categorical variables are processed using CatBoost's native categorical handling. Feature engineering generates additional indicators such as heart-rate-to-age ratios, oxygen-saturation-to-heart-rate ratios, aggregate statistics, and missing value indicators to

capture physiological and behavioral patterns. A CatBoost gradient boosting classifier is then applied to estimate dementia risk from lifestyle and physiological indicators [23], with hyperparameter optimization performed using the Optuna framework and model evaluation conducted using Repeated Stratified K-Fold cross-validation. The model produces a probability score mapped into three categories: Low Risk, At Risk, and High Risk, achieving an AUC of 0.9892 and an accuracy of 94.9%.

Cognitive assessment follows ACE-III scoring [21] across attention, memory, fluency, language, and visuospatial domains to produce a total score out of 100, which is classified as Normal (≥ 88), At Risk (82–87), or Declining (< 82). Functional ability is evaluated using the Lawton IADL scale [22], where female assessments cover eight domains and male assessments five domains, producing functional states categorized as Independent, Partially Dependent, or Dependent. To maintain transparency and clinical interpretability,

the final screening result is determined through a rule-based fusion mechanism combining the wearable physiological risk state derived from the CatBoost model, the cognitive state from ACE-III [21] assessment, and the functional dependency state from Lawton IADL [22] scoring. These indicators are mapped into final dementia risk categories including Normal, Very Mild, Mild, Moderate, and Severe. The outputs from wearable risk prediction, cognitive assessment, and functional evaluation are consolidated into a comprehensive screening report that presents the predicted risk category and supporting indicators, which is delivered to the clinician dashboard within the CogniCare platform to support medical review and further evaluation if required.

B. Handwriting-Based Dementia Screening

The handwriting module analyzes digital writing behavior to identify motor and cognitive patterns associated with early dementia, as changes in handwriting dynamics may reflect impairments in memory, motor coordination, and executive control. The module was developed using the DARWIN handwriting dataset [25], which contains digital handwriting recordings from individuals with and without cognitive impairment represented as sequences of spatial coordinates and timestamps captured during writing and drawing tasks. Within the CogniCare platform, handwriting data are collected through a browser-based drawing interface where users perform predefined tasks while the system records x-y coordinates and timestamps of pen movements. From these trajectories, dynamic behavioral features such as stroke speed, acceleration, movement duration, and inter-stroke timing are extracted to characterize handwriting rhythm and performance. To ensure compatibility with browser-based devices, sensor-dependent variables, including pressure, tilt, azimuth, and in-air signals, were removed. After preprocessing, **40 handwriting features** were retained for model training, with missing values handled using **median imputation** and features standardized using **z-score normalization**. Machine learning models including **Random Forest** and **XGBoost**, were trained to capture non-linear relationships between handwriting dynamics and cognitive status. **SHAP (SHapley Additive Explanations)** was applied to analyze feature importance and identify handwriting characteristics contributing to cognitive impairment predictions.

C. Dementia screening through guided speech-task analysis

The speech module is designed to identify early cognitive risk through structured, guided speech elicitation, as speech production reflects memory retrieval, lexical access, executive control, and motor coordination. Speech data are collected through three guided tasks-self-introduction, morning recall, and picture description-selected to capture complementary domains including verbal fluency, episodic memory, and narrative coherence. Audio responses are recorded through a browser-based interface and stored with session and language metadata.

Each task undergoes dual-pipeline processing. The acoustic model is based on a fine-tuned Wav2Vec2-XLSR-300M architecture, a transformer-based speech representation model trained on large-scale multilingual data. In the acoustic pipeline, speech is resampled to 16 kHz and processed using a fine-tuned **Wav2Vec2-XLSR-300M** model to extract high-level embeddings, along with acoustic indicators such as duration, tempo, zero-crossing rate (ZCR), and root mean square (RMS) energy to characterize speech rhythm and fluency. In the transcript pipeline, recordings are automatically transcribed, and linguistic features including word count, sentence count, speech rate, and lexical variation are extracted to capture memory-related language organization and narrative structure. Task-level probabilities are aggregated using a rule-based fusion mechanism to enhance interpretability. Outputs are mapped into clinically meaningful domain states-fluency, memory-related language, and narrative coherence-and combined into a final speech score (0–100), categorized as Normal, Mild, Moderate, or Severe risk using predefined thresholds. The module generates task-level outputs, domain states, and clinician-facing justification text to ensure transparency, functioning strictly as a screening decision-support tool rather than a diagnostic system.

D. MRI-Based Dementia Stage Classification

The MRI-based dementia classification module analyses structural brain MRI scans to predict dementia severity and support clinical decision-making. The system performs four-class classification consisting of Non-Demented, Very Mild Demented, Mild Demented, and Moderate Demented stages.

The models were trained using T1-weighted structural MRI scans obtained from publicly available datasets like OASIS, where images are labelled using clinically verified diagnoses or Clinical Dementia Rating (CDR) scores. The MRI volumes were pre-processed through skull stripping, intensity normalization, and spatial alignment. Each 3D scan was converted into 2D slices suitable for convolutional neural networks, resized to **224 × 224 pixels**, and normalized using ImageNet statistics. Three convolutional neural network architectures **DenseNet-121**, **EfficientNet-B0**, and **ResNet-50** were used for dementia stage classification. The original classification layers were replaced with task-specific fully connected layers and a four-class Softmax output layer. Models were trained using **cross-entropy loss** and the **Adam optimizer**, with data augmentation techniques such as random rotations and horizontal flips applied to improve generalization. Early stopping based on validation performance was used to reduce overfitting. The outputs of the three models were combined using a **weighted ensemble approach**, where Softmax probabilities were aggregated based on validation performance. To enhance interpretability,

Grad-CAM++ was applied to generate heatmaps highlighting brain regions influencing model predictions.

Within the CogniCare platform, MRI classification is used as an optional advanced assessment when imaging data is available. The predicted stage and confidence scores are integrated into the clinician dashboard alongside results from lifestyle monitoring, cognitive assessments, handwriting analysis, and speech evaluation to support comprehensive dementia risk screening.

IV. IMPLEMENTATION

The CogniCare platform was implemented as a web-based system integrating modules for dementia screening and risk analysis. The frontend, developed using React.js, enables users to perform handwriting tasks, speech recordings, and cognitive assessments through a web browser, while clinicians can upload MRI scans and review screening results. The backend, built with Node.js and Express.js, manages application logic, authentication, and communication with machine learning services and external APIs. Machine learning components were implemented in Python to support multimodal analysis. The Smart Health Monitoring module uses a CatBoost classifier to estimate dementia risk from lifestyle indicators, wearable health data, cognitive assessments, and functional ability scores. The handwriting module analyzes digital writing trajectories using Random Forest and XGBoost, while the speech module extracts acoustic and linguistic features from guided recordings to generate a speech-based cognitive risk score. For neuroimaging analysis, ResNet-50, DenseNet-121, and EfficientNet-B0 were combined in an ensemble model to classify dementia stages, with Grad-CAM highlighting brain regions influencing predictions. All multimodal data and assessment results are stored in MongoDB, while integrations including Google Fit API, Web Speech API, Free Dictionary API, and Ninjas Animal API support wearable data retrieval, speech capture, and verbal fluency validation.

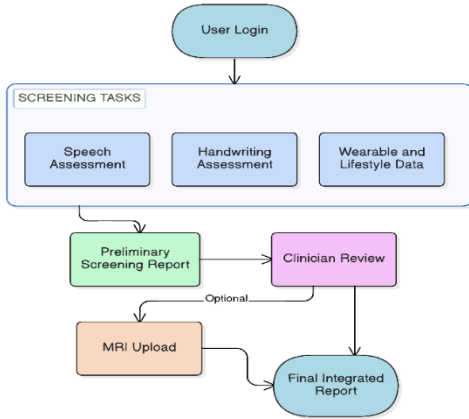


Fig. 1. System Flow

V. RESULTS & EVALUATION

The effectiveness of the proposed CogniCare system was evaluated using standard machine learning metrics including Accuracy, Precision, Recall, F1-score, and Area Under the ROC Curve (AUC). These metrics measure the model's ability to correctly identify dementia-related patterns.

The evaluation was conducted component-wise, analyzing the performance of the wearable-based dementia risk prediction model, handwriting analysis model, Speech analysis model, and MRI analysis model. The results demonstrate that the selected machine learning approaches provide reliable performance for early dementia risk screening.

A. Smart Health Monitoring & Dementia Screening Risk Analysis

The Smart Health Monitoring module uses a CatBoost gradient boosting classifier to estimate dementia risk based on physiological and lifestyle indicators. The model was trained in the processed clinical dataset using stratified cross-validation.

The results demonstrate strong predictive performance, achieving an accuracy of 94.9% and an AUC score of 0.989, indicating effective discrimination between dementia and non-dementia cases. The confusion matrix further confirms that the model correctly classifies most cases, highlighting the suitability of CatBoost for heterogeneous healthcare data.

TABLE I. PERFORMANCE METRICS OF CATBOOST

Metric	Value
Accuracy	0.949
Precision	0.95
Recall	0.95
F1 Score	0.95
AUC	0.989

B. Handwriting-Based Dementia Screening

The handwriting analysis module evaluates writing dynamics to detect cognitive impairment patterns. Multiple feature configurations and machine learning algorithms were evaluated to determine the most effective model for handwriting-based dementia screening.

TABLE II. HANDWRITING MODEL PERFORMANCE

Model Version	Features	Models Trained	Best Model	Best F1	AUC
Full Model	452	Random Forest + XGBoost	XGBoost	0.886	0.950
Browser Model	96	Random Forest + XGBoost	XGBoost	0.869	0.942
Reduced Model	40	Random Forest + XGBoost	Random Forest	0.870	0.932
Ultra Reduced	24	Random Forest + XGBoost	Random Forest	0.838	0.934

Two classification algorithms, **Random Forest** and **XGBoost**, were evaluated using different feature sets, including full, browser-based, and reduced configurations. As shown in Table II, the **40-feature reduced model** achieved a competitive **F1-score of 0.870** while significantly reducing feature dimensionality. Therefore, this

configuration was selected as the final handwriting classification model due to its balance between performance and computational efficiency.

C. Dementia screening through guided speech-task analysis

The speech-based dementia screening module analyzes guided speech responses to detect cognitive impairment. Three models were explored: a **Wav2Vec2-XLSR-300M** deep learning model for raw audio classification, an acoustic feature model using Random Forest, and a transcript-based model using **XGBoost**. In addition, a hybrid fusion model combining Wav2Vec2-XLSR-300M and transcript features was implemented.

TABLE III. PERFORMANCE METRICS OF SPEECH

Model	Accuracy	Precision	Recall	F1 Score	AUC
Wav2Vec2-XLSR-300M	0.79	0.80	0.77	0.78	0.85
Acoustic Model (Random Forest)	0.73	0.74	0.70	0.71	0.79
Transcript Model (XGBoost)	0.76	0.77	0.74	0.75	0.84
Hybrid Fusion Model	0.79	0.81	0.79	0.78	0.85

The results show that the hybrid fusion model achieved the best performance, demonstrating that combining Wav2Vec2-XLSR-300M and linguistic speech features improves dementia risk classification.

D. MRI-Based Dementia Stage Classification

The MRI analysis component classifies dementia stages from brain MRI images using deep learning models. Several convolutional neural network (CNN) architectures were evaluated to identify the most reliable model for dementia stage prediction.

Among the tested models, **DenseNet-121**, **EfficientNet-B0**, and **ResNet-50** showed the best performance. To further improve prediction reliability, an **ensemble model** was developed by combining the outputs of these three networks using a weighted averaging approach.

TABLE IV. PERFORMANCE OF SELECTED MRI CLASSIFICATION MODELS

Model	Accuracy	Macro F1	Macro Precision	Macro Recall
DenseNet-121	0.9484	0.9657	0.9633	0.9684
EfficientNet-B0	0.8562	0.9067	0.9085	0.9171
ResNet-50	0.8406	0.8921	0.8957	0.8889

Ensemble Model	0.9375	0.9606	0.9591	0.9636
-----------------------	---------------	---------------	---------------	---------------

The results indicate that the ensemble model achieved more balanced and robust performance compared to individual models, improving the reliability of dementia stage classification.

VI. DISCUSSION AND CONCLUSION

This study introduced CogniCare, a multimodal AI-assisted dementia screening platform that integrates wearable health monitoring, cognitive assessments, handwriting analysis, speech-based evaluation, and MRI-based stage classification within a unified framework.

Experimental results demonstrate promising performance across the system components. The Smart Health Monitoring module achieved strong predictive capability using a CatBoost model with an AUC of 0.989 and 94.9% accuracy for dementia risk estimation. The handwriting analysis module showed that writing dynamics can indicate cognitive impairment, where a reduced-feature Random Forest model achieved an F1-score of 0.870. In MRI-based analysis, deep learning architectures including DenseNet-121, EfficientNet-B0, and ResNet-50 were evaluated, and the ensemble model achieved 93.75% accuracy with a Macro F1-score of 0.9606.

By combining outputs from multiple modalities, the proposed system improves screening robustness and provides interpretable reports that support clinicians in early dementia risk assessment. CogniCare is designed as a decision-support screening tool, ensuring that final diagnostic decisions remain under medical supervision.

VII. FUTURE WORK

Although the proposed CogniCare platform demonstrates promising results, several opportunities remain for further improvement. Future work will focus on extending the multimodal framework with additional data sources and improving model generalization across diverse populations. One potential direction is the integration of longitudinal monitoring, where wearable indicators, cognitive performance, and behavioral patterns are tracked over time to detect subtle changes in cognitive health. Future research may also explore deep learning-based multimodal fusion models to better capture relationships between modalities such as speech, handwriting, physiological signals, and MRI features.

Further studies will involve clinical validation using larger real-world datasets from healthcare institutions to evaluate the system's reliability in practical settings. In addition, integrating advanced explainable AI techniques and developing mobile-based screening applications could enhance transparency and enable remote cognitive health monitoring.

REFERENCES

- [1] World Health Organization, "Dementia," 2023. [Online]. Available: <https://www.who.int/news-room/fact-sheets/detail/dementia>.
- [2] G. Livingston and J. Huntley, "Dementia prevention, intervention, and care: 2020 report of the Lancet Commission," *Lancet*, vol. 396, no. 10248, pp. 413-446, 2020.
- [3] Alzheimer's Association, "2023 Alzheimer's disease facts and figures," 2023. [Online]. Available: <https://www.alz.org/alzheimers-dementia/facts-figures>.
- [4] N. Bijlani, R. Nilforooshan and S. Kouchaki, "An Unsupervised Data-Driven Anomaly Detection Approach for Adverse Health Conditions in People Living With Dementia: Cohort Study," *JMIR Aging*, vol. 5, no. 1, p. e36580, 2022.
- [5] U. & R. S. Mitra, "ML-powered handwriting analysis for early detection of Alzheimer's disease.," *IEEE Access*, p. 69031–69050, 2024.
- [6] L. Tóth and I. Hoffmann, "A speech recognition-based solution for the automatic detection of mild cognitive impairment from spontaneous speech," *Curr Alzheimer Res*, vol. 15, no. 2, pp. 130-138, 2018.
- [7] C. G. J. Newman, K. Harkness, K. A. Ellis and D. Ames, "Improving the quality of cognitive screening assessments: ACEmobile, an iPad-based version of the Addenbrooke's Cognitive Examination-III," *Alzheimer's & Dementia: Diagnosis, Assessment & Disease Monitoring*, p. 603–611, 2018.
- [8] L.-F. R. O. G.-G. Marco Antonio Esque, *The Internet of Things in dementia: A systematic review*.
- [9] N. A. Albers and e. al., "Wearable multimodal sensors for the detection of behavioral and psychological symptoms of dementia using personalized machine learning models," *Alzheimer's & Dementia: Diagnosis, Assessment & Disease Monitoring*, 2022.
- [10] A. Doherty and e. al., "Daily-life walking speed, running duration and bedtime from wrist-worn sensors predict incident dementia," *International Psychogeriatrics*, 2024.
- [11] A. M. T. V. de Araújo and e. al., "A handwriting-based protocol for assessing neurodegenerative dementia," *Cognitive Computation*, vol. 12, pp. 13-24, 2020.
- [12] Y. Zhang and e. al., "DCDT: A digital clock drawing test system for cognitive impairment screening," in *IEEE Engineering in Medicine and Biology Society (EMBC)*, 2020.
- [13] P. Drotár and e. al., "Handwriting analysis to support neurodegenerative diseases diagnosis: A review," *Pattern Recognition Letters*, vol. 121, pp. 37-45, 2019.
- [14] M. T. P. de Paula and e. al., "Automated free speech analysis reveals distinct markers of Alzheimer's and frontotemporal dementia," *PLOS ONE*, vol. 19, no. 6, 2024.
- [15] S. Luz, F. Haider, S. De La Fuente, D. Fromm and B. MacWhinney, "Alzheimer's Dementia Recognition Through Spontaneous Speech: The ADReSS Challenge," in *INTERSPEECH*, 2020.
- [16] A. König and e. al., "Correlating natural language processing and automated speech analysis with clinician assessment to quantify speech-language changes in mild cognitive impairment and Alzheimer's dementia," *Alzheimer's Research & Therapy*, vol. 13, 2021.
- [17] K. Bäckström, M. Nazari and e. al., "A practical Alzheimer disease classifier via brain imaging-based deep learning on 85,721 samples," *bioRxiv*, 2020.
- [18] C. Salvatore, A. Cerasa and I. Castiglioni, "Using machine learning to quantify structural MRI neurodegeneration patterns of Alzheimer's disease into dementia score," *Scientific Reports*, 2020.
- [19] M. F. M. T. B. Abdullah and e. al., "Generalizable deep learning model for early Alzheimer's disease detection from structural MRIs," *Scientific Reports*, 2022.
- [20] M. K. D. G. Michael David, *Remote Monitoring of Physiology in People Living With Dementia: An Observational Cohort Study*.
- [21] "Addenbrooke's Cognitive Examination — III (ACE-III)," [Online]. Available: <https://camcops.readthedocs.io/en/latest/tasks/ace3.html>.
- [22] M. A. B. Carla Graf, *The Lawton Instrumental Activities of Daily Living (IADL) Scale*.
- [23] Y.-M. C. Y.-H. Y. V. S. T. Benny Wei-Yun Hsu, *Predicting Fine-grained Behavioral and Psychological Symptoms of Dementia Based on Machine Learning and Smart Wearable Device*.
- [24] K. X. S. L. W. L. K. A. L. Y. L. K. X. L. G. Y. L. L. C. M. A. B. & F. L. Ye, "The role of lifestyle factors in cognitive health and dementia in oldest-old: A systematic review," *Neuroscience & Biobehavioral Reviews*, 2023.
- [25] N. D. Cilia, "An experimental protocol to support cognitive impairment diagnosis using handwriting analysis: DARWIN dataset," *Procedia Computer Science*, vol. 136, pp. 370-377, 2018.
- [26] A. Z. , A. M. , S. S. , S. T. A. , T. E. , M. H. , A. A. , M. K. , S. K. , H. R. , R. N. , P. B. Shirin Enshaeifar, *Health management and pattern analysis of daily living activities of people with dementia using in-home sensors and machine learning techniques*.